Experience in the weights:

distilling a frontier agent's continual-learning memory into a small model, and what the memory still does at inference

SenseLab

https://sense-lab.ai

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Abstract

We ran the pipeline WikiSkill describes, with one change at the end: the skill layer is a model. Two frontier agents (gpt-5.5 and claude-opus-5) worked two operations scenarios, a support desk and a CI-fix queue, for 60 episodes each on a memory layer whose entries gain or lose confidence with the outcomes of the decisions that read them, through a regime change at episode 20 where the right fix for some task classes silently changes. Memory is worth a lot to the frontier on its own: 0.12 more success and 0.19 more first-try success than the same agent with no memory, most of it after the change. We then trained SenseLab-tuned Managed Models on the frontier's sealed decision traces, one per scenario and teacher, and served them through a gateway that retrieves from the memory the teacher built. Under the change the teacher lived through, the student with memory reaches 0.94 and 0.92 success where the frontier with memory reaches 0.84 and 0.82, at about one fortieth of the list price per solved task. Under a change no training trace contains, the student matches the frontier on success within noise and recovers about seven tenths of the frontier's first-try gain from memory. Memory still does work at inference for a tuned model: switching it off costs 0.26 first-try success after the change, and serving the context without the outcome record costs 0.24. A skill file maintained the WikiSkill way, with a validation gate, sits 0.10 below the memory layer after the change and costs more per solved task than the frontier on memory. A student trained on its own traces, with no frontier in the loop, improves on its parent. The order matters: the frontier agent on memory is where an account starts and what produces every trace; the tuned model is a later step for a decision that has volume, and the frontier stays in the loop as teacher and fallback.

1 The question

WikiSkill [1] argues that an agent's experience should live in three layers: the raw record of what happened, a curated body of knowledge, and a procedural skill file the agent reads before every task. The paper shows a frontier model carrying its task performance forward through the skill file alone. The skill layer is text, and the model that reads it is the same frontier model that produced the experience, at the same price per token, for every task.

We wanted to know what happens one step further on. A frontier agent running on a continual-learning memory layer accumulates more than a skill file. Its store carries entries whose confidence goes up or down with the outcomes of the decisions that read them, an action record that says which of the available actions worked here and which did not, and a sealed decision trace for every task (the request, what was retrieved, what the agent did, how it turned out). That trace is a supervised example. So: can the experience a frontier agent accumulates be distilled into a model an order of magnitude smaller, how close does that student come to the frontier agent it learned from, and once the experience is in the weights, does the memory layer still do anything useful at inference?

SenseLab already keeps the three layers. Decision traces are the raw layer; the memory, with its outcome-reinforced confidence and action record, is the wiki; and *Managed Models*, tuned on the account's own traces and served through an inference gateway that retrieves from the same memory the teacher used, are the skill layer in weights. This paper is that pipeline run end to end on a benchmark whose ground truth we control, with the frontier teacher measured on the same held-out seeds, under the same harness, at the same list prices. The study is preregistered [3]. The harness, the training matrix, the dataset snapshots and every per-episode record are published with it.

The order of operations. Nothing here starts with a small model. The first step, and the one every account takes, is a frontier agent running on the memory layer. That is where we measure what memory is worth to a frontier agent (`F1` against `F0` in §4), and it is also the step that produces everything a student is made of: the sealed traces are its training set, and the store it is later served from was built by the frontier agent's reads, writes and outcomes. Distillation comes after that store exists, once some decision has become frequent enough that its cost matters. At that point the frontier agent has learned the job on SenseLab, and a Managed Model is trained to make that one decision at a fraction of the price, served with the same memory. The frontier does not leave the loop. It is still the teacher for the next scenario and the next regime, it still handles every task outside a student's scope, and the gateway can route an unusable student answer back to it (we switched that off here so that a student's failures are its own). What we are asking is how much of the frontier agent's experience survives the hand-off, and what the memory still does for the student afterwards.

2 What is being compared

The benchmark. We reuse the continual-learning benchmark of our companion study [2]. An operations agent works a stream of tasks that never repeat exactly (support tickets, CI failures) against a store pre-loaded with 300 realistic distractor notes, in a world whose rules change at episode 20, when half the task classes acquire a new correct action. Sixty episodes, run both as a single agent and as a fleet of six sharing one store, with a retry budget of three. Every result below is on three seeds that no teacher cell used. Task identifiers (ticket and PR numbers) are drawn at random per task; §3.1 explains why we have to say so.

Metrics. *Success* means the task was solved within the retry budget; *first-try* means it was solved on the first attempt; *escalated* means the budget was spent without a solution. The budget matters for how the numbers read. An agent that knows the handful of plausible fixes for a task class can reach the right one on a second attempt with no memory at all, so success saturates for such an agent. What memory buys then shows up as first-try success (the decision made before any feedback) and as fewer attempts per solved task. We report both everywhere and give the paired tests on both.

Teachers. `F1` is a frontier model (GPT-5.5 or Claude Opus 5) running the full tool loop on the SenseLab arm: a briefing, retrieval with the outcome record, domain and diagnostic tools, a reflection note after each episode, and an outcome commit that credits the memories it read. `F0` is the same model with no memory. The teacher grid produced 720 sealed traces per (scenario, teacher model), of which 81 to 87% are successes; the rest are labelled failures with the failing attempt marked.

Students. A student is a SenseLab Managed Model of the `senselab-s1` family: a small model that SenseLab fine-tunes through its own pipeline on the traces of one teacher's cells, and on nothing else in the account. Scoping is by the cells' memory paths (§3). A training row is what the product's renderer produced for the teacher: the task, the memory context the teacher saw, the actions already tried in that task and how they fared, and the outcome record's recommendation. The target is the teacher's last consequential action. The student is served by the inference gateway, which retrieves from the *evaluation* cell's own store, renders the same prompt shape, and seals the decision as a trace. One action per call. We trained ten students. For each scenario there is a GPT-5.5 student and a Claude Opus 5 student, each in two generations: *v1* on the teacher grid of the companion study and *v2* on a fresh teacher grid run after the fix in §3.1. For the support scenario there is also a student of a teacher that was served without action priors. We treat the generations as replicates; every student was evaluated on the same held-out seeds.

Decision mode. The student never runs the tool loop. The harness puts the task to the gateway, an action comes back, and the environment executes it. A failed attempt goes back as a failed tool result, which the gateway renders as *Tried in this task*, the same shape a fail-then-succeed teacher trace was trained on. Diagnostic tools are not available, so a diagnostic answer counts as a failed attempt. The student writes no reflection of its own.

Conditions. Table 1 defines them. The letter says whose weights answer. F is the frontier model running the full agent loop; D is the frontier model answering the student's rendered prompt in one shot; S is the *student*, the SenseLab Managed Model fine-tuned on the teacher's traces; B is the same Managed Model before tuning. The digit says whether SenseLab is live at inference: under 1 the condition reads from and writes to a memory store that fills over the run, under 0 the store is empty and nothing is served. A suffix narrows what the live store serves to the student.

id	whose weights	memory at inference, and what the condition isolates
F1	frontier, full agent loop	SenseLab live; the teacher on held-out seeds (the bar)
F0	frontier, full agent loop	no memory
D1	frontier, one shot	the student's rendered prompt, SenseLab live; the prompt's worth
D0	frontier, one shot	the student's rendered prompt, empty store
S1	SenseLab-tuned Managed Model	SenseLab live: context, avoid rows, action priors; the claim
S1n	SenseLab-tuned Managed Model	S1 plus a GPT-5.4-mini note-writer after each episode
S1p	SenseLab-tuned Managed Model	context only, no avoid rows, no priors
S1k	SenseLab-tuned Managed Model	store present, nothing served; the weights alone
S0	SenseLab-tuned Managed Model	empty store; the weights alone
B1	Managed Model base, untuned	SenseLab live on the same prompt; what tuning added

Table 1: Conditions. Every condition runs the same scenarios, seeds, episodes, regime change, distractor load and retry budget; students and controls run in decision mode.

In more detail:

- S1 is the SenseLab-tuned Managed Model, served by the inference gateway with the full memory it was trained to read: retrieved context, avoid rows for entries that have been discredited here, and the action record with its recommendation. The memory is the evaluation cell's own. It

starts empty and is written by the outcomes the harness commits, the same way the teacher's was. This is the condition the paper is about.

- S0 is the same tuned weights with an empty store, so nothing is retrieved. Everything S0 knows is in the weights. S1−S0 is what the memory is worth at inference on top of the experience already distilled.
- S1k is the tuned weights with the store present and written to, and nothing served (a context of zero entries). It checks that S0 is not an artefact of the empty store.
- S1p is the tuned weights served the memory context without the outcome record (no avoid rows, no action priors). S1−S1p is what the outcome record adds.
- S1n is S1 plus a cheap note-writer (GPT-5.4-mini) that reflects after each episode, so the store keeps learning from the student's own outcomes the way the teacher's did from its reflections.
- B1 is the Managed Model's base weights before tuning, on the identical served prompt with live memory. S1−B1 is what tuning added.
- F1 and F0 are the frontier teacher (GPT-5.5 or Claude Opus 5) running the full tool loop, with SenseLab live (F1, the bar we measure the student against) and with no memory (F0).
- D1 and D0 are the frontier model given the identical rendered prompt the student receives (the gateway's `render_only` mode returns the bytes and sends nothing upstream), with live memory (D1) and on an empty store (D0). D1−S1 is what the frontier's weights add to the same prompt, and F1−D1 is what the tool loop and the agent's own reflection add.

A skill-file arm. To put the WikiSkill design on the same axes we add `skillfile`, a small but faithful implementation. It keeps raw episode records. A maintainer call every five episodes folds them into a wiki (`patterns.md`). A proposer drafts `SKILL.md` from the wiki, the current file and the impact log, and the draft is accepted only if a validation replay on regenerated instances (another seed, same regime) does at least as well as the current file, and strictly better when one exists. The file is prepended to the system prompt; the arm has no memory tools. Its maintainer, proposer and validation calls are charged to the arm. Every episode is also sealed as a SenseLab trace, so the skill-file agent can be a teacher too (K1 is the arm, K0 its student).

3 Method

Scoping a student to a teacher. A Managed Model learns from the account's traces that fall in its scope. The harness names each cell's memory by a path unique to the cell (`clbench/support-senselab-gpt-5-5-s11-...`). Every trace the cell seals carries that path on the entries it read and, since this study, as a trace attribute. A student's scope is the list of its teacher cells' paths (`entity_prefix`, a |-separated list, matched exactly or as a parent). We checked the predicate against the production store before training: for the v1 students, whose teacher traces predate the scope attribute, the predicate selected 632 to 696 of each teacher's 660 to 720 sealed traces (the rest are episodes in which the agent read nothing, so no entry carried the cell's path); for the v2 students, whose traces carry the scope as an attribute, 705 to 720 of 720. Every selected trace had a target call and an outcome, and the product's readiness check found 8 task shapes and a positive share of 0.74 to 0.90 in every scope.

What a training row contains. The renderer that builds training rows is the renderer that serves. A row has the task text; up to three memory entries with their evidence status; the actions already tried in this task with their result; the action record's recommendation when one exists; and the target, the last non-exploratory tool call as a JSON object. Fail-then-succeed traces yield a row per attempt boundary, so the student sees "tried X, failed" and then the teacher's next choice. Rows are split 70/15/15 into tuning, tuning-validation and a held-out evaluation set that the product scores every version on: tool agreement, decision agreement (tool plus the action argument), exact argument agreement, parse rate, failure reproduction, and stability under rewording. We publish the dataset hashes and the rows themselves.

The gate. The product refuses to promote a version that fails its default bars (tool agreement ≥ 0.75, exact agreement ≥ 0.45, parse rate ≥ 0.90). A student that fails the gate is a finding in itself, so the study promotes under relaxed bars and reports the default verdict beside every student.

Serving. Per-model serving options fix what the prompt contains: context entries, avoid rows, priors, the retrieval path, and whether an unusable answer falls back to a frontier model. Fallback is off for every student here, so a student's failures are its own. The ablations `S1p` and `S1k` are the same weights served narrower through a per-request option the gateway only honours in the tightening direction.

Reading an answer. Decision-mode answers are one JSON object naming a tool and its arguments. The untuned base tends to write the decision key where the tool name goes (`resolve:escalate_tier2`), or the bare action (`refund`). The harness reads both as the terminal tool with that action, for every condition that answers in this form, so a model is graded on the decision it made and not on where it put the colon. A diagnostic tool named in place of a decision (`get_pr_commits`) stays what it is: a non-terminal answer, counted as a failed attempt.

Cost. We use list prices for every model (§7) and report the provider cost SenseLab pays for a student separately. The note-writer's calls and the skill-file arm's maintenance calls are charged to their conditions.

3.1 A shortcut the first student found, and the fix

The first student we trained scored 0.81 post-change success on the changed task classes with no memory at all (a store it had never written to and read nothing from), where the frontier model without memory scores 0.04. Nothing in the weights should know that the world changed at episode 20. Something did. The benchmark's ticket and PR numbers were counted from the episode index (`SUP-4400+ep`, `PR #3000+7ep+seed`), and since the regime change always falls at the same episode, the identifier encoded the regime. The student had learned "`SUP-442x` and up means the new fix" from six seeds of teacher traces. No memory layer is needed for that shortcut, and no human agent could use it.

The fix is a scenario flag (`opaque_ids`) that draws identifiers from the task instance's own random stream. Every distillation cell runs with it. Every run before the fix is archived under a `leaky-` prefix and appears in no table of this paper. The v1 students keep the weights they learned from counted identifiers and are evaluated where the shortcut cannot fire; the v2 students were trained on a fresh teacher grid run on the fixed scenario, so the two generations differ only

in whether the artefact was in their training data. We report the episode because a distillation pipeline will find this kind of thing in any benchmark whose surface form correlates with its hidden state, and because a memory paper that had missed it would have credited the memory with the student's clairvoyance.

3.2 A change the weights have never seen

Under the benchmark's regime change, every post-change (task class, correct action) pair occurs in the students' training traces: the teachers lived through the change, and their traces from episodes 20 to 60 show the new fix winning. A student trained on those traces carries both regimes in its weights. What it needs memory for is to tell which regime the current task is in, and as §4 shows, a retry budget of three lets it find out by trying. That makes the trained-on change a weak test of memory at inference.

So we added a second change set, B, and ran the students under it with nothing retrained. B changes classes that A leaves alone (three of eight in support, two of eight in CI) to actions that no A episode ever pairs them with, following A's own design: in support, two quirk-to-quirk changes and one change to the textbook answer; in CI, one quirk-to-textbook and one textbook-to-quirk change. The pre-change world is identical to A's. Under B nothing in a student's weights knows the new fix, so whatever it recovers after episode 20 is what its memory gives it. We wrote the preregistration's amendment (H8) when 77 frontier and 17 student episodes of the B grid existed and no student-with-memory rows. It predicts that `S0`'s first-try success on the changed classes after the change falls to `F0`'s, and that `S1` recovers above `S0` by at least the gap measured under A.

4 Results

Tables 2 and 3 give every condition on the held-out seeds (101, 113, 127; fleets of 1 and 6; 60 episodes; models pooled per condition), and Tables 4 and 5 give the paired deltas the hypotheses are stated in, over all episodes, before the change, and after it.

Step one is the frontier on memory. `F1` over `F0` is +0.12 success and +0.19 first-try success over the run; before the change the two are level on success and memory already adds 0.11 first-try, after it memory adds 0.19 success and 0.23 first-try. On the changed support classes the frontier without memory recovers to 0.04; with memory it reaches 0.55. That gain comes with no training and nothing changed in the agent beyond the memory tools, and every trace a student learns from below was sealed while it was being earned.

The students learned the job. `S1`, the SenseLab-tuned Managed Model served with the same memory, is at 0.94 success on CI-fix and 0.92 on support, against 0.84 and 0.82 for the frontier on memory, and it escalates a third as often. Paired by task, `S1` is +0.22 over `F0` where the teacher's own gain was +0.12 (H1 holds with room to spare), at $0.0014 per solved task against $0.061 for `F1` at list prices. We are careful about what this says: under change set A the students' training traces contain both regimes, so a student knows the post-change fixes in its weights and the frontier has to learn them from memory during the run. §5 takes that advantage away.

Memory does work at inference even so. `S1` over `S0` is +0.05 success and +0.12 first-try; split around the change it is −0.03 and −0.15 before, +0.09 and +0.26 after (H2). `S1k`, which has a store present but is served nothing, is indistinguishable from `S0`, so the effect is the served context and not the presence of the store. Serving the context without the outcome record (`S1p`, no avoid rows and no priors) gives back most of that: `S1` over `S1p` is +0.24 first-try after the change (H3). The pre-change numbers are a real cost and we report them as such: on a young store, before any outcome has been committed, the avoid rows and priors point at the wrong things often enough to

lose 0.15 first-try success. Table 7 shows the same shape for all ten students; the gpt-5.5 students pay more before the change than the claude-opus-5 students and gain about the same after it.

The prompt is not the whole story. The frontier answering the student's exact served prompt (D1) is 0.13 success below the student (H4), so the weights carry something the prompt does not, and the full tool loop with the agent's own reflection (F1 over D1) adds 0.03 success and 0.12 first-try, most of it before the change. The frontier in decision mode with an empty store (D0) escalated every task and is not a usable control; we say more in the threats. A cheap note-writer reflecting after each student episode (S1n) did not help (−0.02, H5 falsified) and costs two and a half times as much. The tuning is what made the student competent: the untuned base behind the same prompt (B1) is 0.49 success below S1 (H6).

The skill file (K1, the frontier reading a SKILL.md that a maintainer, a proposer and a validation gate rebuild every five episodes) is level with the memory layer before the change and 0.10 below it on the changed classes after it (H7). On CI-fix it lands exactly on the frontier with no memory: 714 of 720 tasks end the same way. The impact logs say why: after the change, the proposer's new drafts tie the current file on the four-episode validation replay and the gate rolls them back, so the file never learns the new fixes. The gate is also where the money goes. Counting the maintainer, the proposer and the two replays per checkpoint, K1 costs $0.087 per solved task, above the frontier on memory.

condition	n	success	first-try	escalated	tokens/ep	$/success	pre→post(changed)	unparse/ep
F1 frontier, full loop + SenseLab	720	0.844 [0.81,0.88]	0.753 [0.71,0.79]	0.156 [0.12,0.19]	10735	$0.0618	0.983 → 0.550	0.0
F0 frontier, full loop, no memory	720	0.787 [0.76,0.82]	0.664 [0.63,0.70]	0.212 [0.18,0.24]	2481	$0.0176	0.963 → 0.400	0.0
D1 frontier @decision (served prompt) + SenseLab	1440	0.818 [0.79,0.84]	0.608 [0.58,0.64]	0.182 [0.16,0.21]	1194	$0.0127	0.835 → 0.633	0.0
D0 frontier @decision, empty store	1440	0.001 [0.00,0.00]	0.000 [0.00,0.00]	0.999 [1.00,1.00]	1206	$42.4754	0.000 → 0.000	0.01
S1 student + SenseLab	1440	0.942 [0.91,0.97]	0.794 [0.74,0.85]	0.058 [0.03,0.09]	819	$0.0011	0.971 → 0.858	0.0
S1n student + SenseLab + note-writer	1440	0.925 [0.89,0.96]	0.780 [0.74,0.81]	0.075 [0.04,0.11]	2538	$0.0032	0.914 → 0.879	0.0
S1p student + context only (no avoid/priors)	1440	0.919 [0.88,0.95]	0.742 [0.72,0.77]	0.081 [0.05,0.12]	543	$0.0009	1.000 → 0.758	0.0
S1k student, nothing served (k0)	1440	0.913 [0.87,0.95]	0.722 [0.70,0.75]	0.087 [0.05,0.13]	186	$0.0005	1.000 → 0.742	0.0
S0 student, no memory	5760	0.914 [0.90,0.93]	0.723 [0.71,0.74]	0.086 [0.07,0.10]	186	$0.0005	1.000 → 0.744	0.0
B1 untuned base + SenseLab	360	0.381 [0.27,0.51]	0.247 [0.17,0.34]	0.619 [0.49,0.74]	1754	$0.0005	0.483 → 0.050	0.0
K1 frontier, full loop + skill file	720	0.787 [0.76,0.82]	0.706 [0.67,0.74]	0.212 [0.18,0.24]	14166	$0.0889	0.963 → 0.400	0.0

Table 2: Held-out results, `ci-fix` scenario, fleet sizes pooled.

condition	n	success	first-try	escalated	tokens/ep	$/success	pre→post(changed)	unparse/ep
F1 frontier, full loop + SenseLab	720	0.818 [0.79,0.85]	0.696 [0.67,0.72]	0.182 [0.15,0.21]	9622	$0.0594	0.933 → 0.550	0.0
F0 frontier, full loop, no memory	720	0.629 [0.61,0.65]	0.410 [0.39,0.43]	0.371 [0.35,0.39]	2439	$0.0231	0.958 → 0.042	0.0
D1 frontier @decision (served prompt) + SenseLab	2160	0.780 [0.75,0.81]	0.603 [0.59,0.62]	0.220 [0.19,0.25]	1264	$0.0153	0.850 → 0.551	0.0
D0 frontier @decision, empty store	2160	0.000 [0.00,0.00]	0.000 [0.00,0.00]	1.000 [1.00,1.00]	1297	—	0.000 → 0.000	0.001
S1 student + SenseLab	2160	0.918 [0.90,0.94]	0.747 [0.72,0.77]	0.082 [0.06,0.10]	1030	$0.0016	0.936 → 0.829	0.0
S1n student + SenseLab + note-writer	2160	0.905 [0.89,0.92]	0.746 [0.73,0.76]	0.095 [0.08,0.11]	2667	$0.0037	0.923 → 0.810	0.001
S1p student + context only (no avoid/priors)	2160	0.884 [0.86,0.91]	0.583 [0.56,0.60]	0.116 [0.09,0.14]	692	$0.0014	0.982 → 0.701	0.001
S1k student, nothing served (k0)	2160	0.844 [0.82,0.87]	0.570 [0.55,0.59]	0.156 [0.13,0.18]	296	$0.0010	0.963 → 0.639	0.0
S0 student, no memory	8640	0.843 [0.83,0.86]	0.570 [0.56,0.58]	0.156 [0.14,0.17]	297	$0.0010	0.963 → 0.632	0.0
B1 untuned base + SenseLab	360	0.503 [0.45,0.57]	0.383 [0.34,0.43]	0.497 [0.43,0.55]	1588	$0.0004	0.592 → 0.300	0.006
K1 frontier, full loop + skill file	720	0.743 [0.70,0.78]	0.588 [0.54,0.64]	0.257 [0.22,0.30]	11562	$0.0854	0.963 → 0.296	0.0

Table 3: Held-out results, `support` scenario, fleet sizes pooled.

test	all episodes	pre-change	post-change
H1 student recovers the teacher's gain	+0.222 [+0.18,+0.26] n=720	-0.007 [-0.02,+0.01] n=240	+0.336 [+0.27,+0.40] n=480
teacher's gain (reference)	+0.123 [+0.08,+0.17] n=720	-0.002 [-0.04,+0.03] n=240	+0.185 [+0.12,+0.25] n=480
H2 memory at inference: S1 − S0	+0.051 [+0.04,+0.07] n=720	-0.028 [-0.04,-0.02] n=240	+0.091 [+0.07,+0.12] n=480
H2 memory at inference: S1 − S1k	+0.051 [+0.04,+0.07] n=720	-0.028 [-0.04,-0.02] n=240	+0.091 [+0.07,+0.12] n=480
H3 outcome record: S1 − S1p	+0.028 [+0.02,+0.04] n=720	-0.037 [-0.05,-0.03] n=240	+0.061 [+0.05,+0.07] n=480
H4 frontier weights on the same prompt: D1 − S1	-0.131 [-0.16,-0.10] n=720	-0.111 [-0.17,-0.05] n=240	-0.142 [-0.17,-0.12] n=480
H4 tool loop + own reflection: F1 − D1	+0.032 [+0.00,+0.06] n=720	+0.116 [+0.07,+0.16] n=240	-0.009 [-0.05,+0.03] n=480
H5 note-writer: S1n − S1	-0.015 [-0.03,-0.00] n=720	-0.035 [-0.06,-0.02] n=240	-0.005 [-0.02,+0.01] n=480
H6 tuning over base: S1 − B1	+0.489 [+0.42,+0.57] n=720	+0.416 [+0.34,+0.50] n=240	+0.525 [+0.45,+0.61] n=480
H7 skill file vs memory: K1 − F1	-0.066 [-0.10,-0.04] n=720	+0.004 [-0.03,+0.04] n=240	-0.101 [-0.14,-0.06] n=480

Table 4: Paired success-rate deltas (condition A minus B), 95% bootstrap intervals over cells.

test	all episodes	pre-change	post-change
H1 student recovers the teacher's gain	+0.234 [+0.17,+0.30] n=720	+0.067 [+0.01,+0.12] n=240	+0.317 [+0.23,+0.39] n=480
teacher's gain (reference)	+0.188 [+0.12,+0.25] n=720	+0.110 [+0.08,+0.15] n=240	+0.226 [+0.13,+0.32] n=480
H2 memory at inference: S1 − S0	+0.124 [+0.09,+0.15] n=720	-0.148 [-0.17,-0.12] n=240	+0.260 [+0.21,+0.31] n=480
H2 memory at inference: S1 − S1k	+0.124 [+0.09,+0.15] n=720	-0.149 [-0.17,-0.13] n=240	+0.260 [+0.21,+0.31] n=480
H3 outcome record: S1 − S1p	+0.108 [+0.07,+0.14] n=720	-0.147 [-0.17,-0.12] n=240	+0.236 [+0.19,+0.28] n=480
H4 frontier weights on the same prompt: D1 − S1	-0.166 [-0.18,-0.15] n=720	-0.231 [-0.28,-0.19] n=240	-0.133 [-0.15,-0.12] n=480
H4 tool loop + own reflection: F1 − D1	+0.119 [+0.09,+0.15] n=720	+0.275 [+0.20,+0.35] n=240	+0.042 [+0.01,+0.07] n=480
H5 note-writer: S1n − S1	-0.008 [-0.03,+0.01] n=720	-0.038 [-0.07,-0.01] n=240	+0.007 [-0.01,+0.03] n=480
H6 tuning over base: S1 − B1	+0.455 [+0.39,+0.53] n=720	+0.410 [+0.36,+0.47] n=240	+0.478 [+0.40,+0.56] n=480
H7 skill file vs memory: K1 − F1	-0.078 [-0.11,-0.04] n=720	-0.008 [-0.05,+0.03] n=240	-0.113 [-0.17,-0.06] n=480

Table 5: Paired first-attempt success deltas (condition A minus B), 95% bootstrap intervals over cells.

Table 6: Each student: held-out task success and the product's own evaluation of the promoted version.

model	cond	success	first-try	$/success	dec-agr	parse
ci-base-s1	B1	0.381 [0.27,0.51]	0.247 [0.17,0.34]	$0.0005	—	—
ci-sl-gpt55	S1	0.814 [0.77,0.86]	0.603 [0.53,0.68]	$0.0019	0.9877	1.0
ci-sl-gpt55	S0	0.808 [0.76,0.86]	0.656 [0.61,0.70]	$0.0007	0.9877	1.0
ci-sl-gpt55+notes=gpt-5.4-mini	S1n	0.811 [0.78,0.84]	0.658 [0.62,0.70]	$0.0044	0.9877	1.0
ci-sl-gpt55+notes=gpt-5.4-mini	S0	0.808 [0.76,0.86]	0.656 [0.61,0.70]	$0.0007	0.9877	1.0
ci-sl-gpt55-v2	S1	0.964 [0.94,0.99]	0.792 [0.77,0.82]	$0.0011	0.9406	1.0
ci-sl-gpt55-v2	S0	1.000 [1.00,1.00]	0.761 [0.72,0.80]	$0.0005	0.9406	1.0
ci-sl-gpt55-v2+notes=gpt-5.4-mini	S1n	0.927 [0.89,0.96]	0.804 [0.78,0.83]	$0.0033	0.9406	1.0
ci-sl-gpt55-v2+notes=gpt-5.4-mini	S0	1.000 [1.00,1.00]	0.761 [0.72,0.80]	$0.0005	0.9406	1.0
ci-sl-gpt55-v2~k0	S1k	1.000 [1.00,1.00]	0.761 [0.72,0.80]	$0.0005	0.9406	1.0
ci-sl-gpt55-v2~k0	S0	1.000 [1.00,1.00]	0.761 [0.72,0.80]	$0.0005	0.9406	1.0
ci-sl-gpt55-v2~plain	S1p	1.000 [1.00,1.00]	0.761 [0.72,0.80]	$0.0008	0.9406	1.0
ci-sl-gpt55-v2~plain	S0	1.000 [1.00,1.00]	0.761 [0.72,0.80]	$0.0005	0.9406	1.0
ci-sl-gpt55~k0	S1k	0.803 [0.76,0.84]	0.656 [0.61,0.70]	$0.0007	0.9877	1.0
ci-sl-gpt55~k0	S0	0.800 [0.76,0.84]	0.656 [0.61,0.70]	$0.0007	0.9877	1.0
ci-sl-gpt55~plain	S1p	0.861 [0.82,0.90]	0.683 [0.65,0.71]	$0.0011	0.9877	1.0
ci-sl-gpt55~plain	S0	0.803 [0.76,0.84]	0.656 [0.61,0.70]	$0.0007	0.9877	1.0
ci-sl-opus5	S1	0.992 [0.99,1.00]	0.908 [0.88,0.93]	$0.0008	0.9886	1.0
ci-sl-opus5	S0	0.844 [0.81,0.88]	0.761 [0.72,0.80]	$0.0005	0.9886	1.0
ci-sl-opus5+notes=gpt-5.4-mini	S1n	0.972 [0.93,1.00]	0.824 [0.76,0.88]	$0.0026	0.9886	1.0
ci-sl-opus5+notes=gpt-5.4-mini	S0	0.853 [0.83,0.88]	0.761 [0.72,0.80]	$0.0005	0.9886	1.0
ci-sl-opus5-v2	S1	1.000 [1.00,1.00]	0.875 [0.85,0.90]	$0.0009	0.9773	0.9714
ci-sl-opus5-v2	S0	1.000 [1.00,1.00]	0.719 [0.67,0.77]	$0.0004	0.9773	0.9714
ci-sl-opus5-v2+notes=gpt-5.4-mini	S1n	0.992 [0.97,1.00]	0.833 [0.81,0.86]	$0.0026	0.9773	0.9714
ci-sl-opus5-v2+notes=gpt-5.4-mini	S0	0.997 [0.99,1.00]	0.706 [0.68,0.73]	$0.0004	0.9773	0.9714
ci-sl-opus5-v2~k0	S1k	0.997 [0.99,1.00]	0.714 [0.68,0.75]	$0.0004	0.9773	0.9714
ci-sl-opus5-v2~k0	S0	0.997 [0.99,1.00]	0.714 [0.68,0.75]	$0.0004	0.9773	0.9714

model	cond	success	first-try	$/success	dec-agr	parse
ci-sl-opus5-v2~plain	S1p	1.000 [1.00,1.00]	0.761 [0.72,0.80]	$0.0007	0.9773	0.9714
ci-sl-opus5-v2~plain	S0	1.000 [1.00,1.00]	0.722 [0.69,0.76]	$0.0004	0.9773	0.9714
ci-sl-opus5~k0	S1k	0.852 [0.82,0.88]	0.759 [0.72,0.80]	$0.0005	0.9886	1.0
ci-sl-opus5~k0	S0	0.867 [0.84,0.89]	0.761 [0.72,0.80]	$0.0005	0.9886	1.0
ci-sl-opus5~plain	S1p	0.817 [0.78,0.85]	0.761 [0.72,0.80]	$0.0010	0.9886	1.0
ci-sl-opus5~plain	S0	0.853 [0.83,0.88]	0.761 [0.72,0.80]	$0.0005	0.9886	1.0
sup-base-s1	B1	0.503 [0.45,0.57]	0.383 [0.34,0.43]	$0.0004	—	—
sup-nopriors-gpt55	S1	0.856 [0.82,0.89]	0.653 [0.60,0.70]	$0.0019	0.9851	1.0
sup-nopriors-gpt55	S0	0.828 [0.81,0.84]	0.492 [0.48,0.50]	$0.0011	0.9851	1.0
sup-nopriors-gpt55+notes=gpt-5.4-mini	S1n	0.831 [0.80,0.87]	0.660 [0.63,0.69]	$0.0045	0.9851	1.0
sup-nopriors-gpt55+notes=gpt-5.4-mini	S0	0.833 [0.81,0.86]	0.486 [0.48,0.49]	$0.0011	0.9851	1.0
sup-nopriors-gpt55-v2	S1	0.839 [0.80,0.88]	0.714 [0.70,0.73]	$0.0018	0.9737	1.0
sup-nopriors-gpt55-v2	S0	0.767 [0.74,0.79]	0.594 [0.59,0.61]	$0.0011	0.9737	1.0
sup-nopriors-gpt55-v2+notes=gpt-5.4-mini	S1n	0.881 [0.87,0.89]	0.742 [0.71,0.77]	$0.0038	0.9737	1.0
sup-nopriors-gpt55-v2+notes=gpt-5.4-mini	S0	0.761 [0.72,0.79]	0.600 [0.59,0.61]	$0.0011	0.9737	1.0
sup-nopriors-gpt55-v2~k0	S1k	0.758 [0.73,0.79]	0.594 [0.58,0.61]	$0.0011	0.9737	1.0
sup-nopriors-gpt55-v2~k0	S0	0.742 [0.71,0.77]	0.583 [0.57,0.59]	$0.0011	0.9737	1.0
sup-nopriors-gpt55-v2~plain	S1p	0.825 [0.80,0.84]	0.614 [0.60,0.62]	$0.0014	0.9737	1.0
sup-nopriors-gpt55-v2~plain	S0	0.744 [0.70,0.78]	0.589 [0.58,0.60]	$0.0011	0.9737	1.0
sup-nopriors-gpt55~k0	S1k	0.828 [0.81,0.85]	0.492 [0.48,0.50]	$0.0011	0.9851	1.0
sup-nopriors-gpt55~k0	S0	0.825 [0.81,0.84]	0.489 [0.48,0.50]	$0.0011	0.9851	1.0
sup-nopriors-gpt55~plain	S1p	0.889 [0.87,0.91]	0.514 [0.49,0.53]	$0.0015	0.9851	1.0
sup-nopriors-gpt55~plain	S0	0.822 [0.81,0.84]	0.494 [0.49,0.51]	$0.0011	0.9851	1.0
sup-sl-gpt55	S1	0.983 [0.97,1.00]	0.811 [0.78,0.84]	$0.0012	0.8861	1.0
sup-sl-gpt55	S0	0.908 [0.89,0.92]	0.558 [0.55,0.57]	$0.0009	0.8861	1.0
sup-sl-gpt55+notes=gpt-5.4-mini	S1n	0.969 [0.93,1.00]	0.807 [0.79,0.82]	$0.0029	0.8861	1.0
sup-sl-gpt55+notes=gpt-5.4-mini	S0	0.906 [0.89,0.92]	0.556 [0.54,0.56]	$0.0009	0.8861	1.0
sup-sl-gpt55-v2	S1	0.919 [0.90,0.94]	0.781 [0.74,0.81]	$0.0015	1.0	0.9907
sup-sl-gpt55-v2	S0	0.925 [0.91,0.94]	0.644 [0.64,0.65]	$0.0008	1.0	0.9907
sup-sl-gpt55-v2+notes=gpt-5.4-mini	S1n	0.883 [0.86,0.90]	0.756 [0.74,0.78]	$0.0038	1.0	0.9907
sup-sl-gpt55-v2+notes=gpt-5.4-mini	S0	0.942 [0.93,0.95]	0.644 [0.64,0.65]	$0.0008	1.0	0.9907
sup-sl-gpt55-v2~k0	S1k	0.936 [0.92,0.95]	0.650 [0.65,0.65]	$0.0008	1.0	0.9907
sup-sl-gpt55-v2~k0	S0	0.933 [0.92,0.94]	0.650 [0.65,0.65]	$0.0008	1.0	0.9907
sup-sl-gpt55-v2~plain	S1p	0.956 [0.94,0.97]	0.642 [0.64,0.65]	$0.0011	1.0	0.9907
sup-sl-gpt55-v2~plain	S0	0.933 [0.92,0.95]	0.644 [0.64,0.65]	$0.0008	1.0	0.9907
sup-sl-gpt55~k0	S1k	0.917 [0.91,0.93]	0.558 [0.55,0.57]	$0.0009	0.8861	1.0
sup-sl-gpt55~k0	S0	0.911 [0.90,0.92]	0.561 [0.55,0.57]	$0.0009	0.8861	1.0
sup-sl-gpt55~plain	S1p	0.938 [0.92,0.95]	0.621 [0.60,0.64]	$0.0011	0.8861	1.0
sup-sl-gpt55~plain	S0	0.917 [0.90,0.93]	0.556 [0.55,0.56]	$0.0009	0.8861	1.0
sup-sl-opus5	S1	0.928 [0.91,0.94]	0.750 [0.72,0.78]	$0.0017	0.9634	1.0
sup-sl-opus5	S0	0.747 [0.74,0.76]	0.511 [0.50,0.52]	$0.0014	0.9634	1.0
sup-sl-opus5+notes=gpt-5.4-mini	S1n	0.914 [0.90,0.93]	0.725 [0.69,0.76]	$0.0039	0.9634	1.0
sup-sl-opus5+notes=gpt-5.4-mini	S0	0.742 [0.74,0.75]	0.514 [0.51,0.52]	$0.0014	0.9634	1.0
sup-sl-opus5-v2	S1	0.983 [0.97,0.99]	0.772 [0.75,0.79]	$0.0015	0.9545	1.0
sup-sl-opus5-v2	S0	0.894 [0.88,0.91]	0.617 [0.61,0.63]	$0.0010	0.9545	1.0
sup-sl-opus5-v2+notes=gpt-5.4-mini	S1n	0.953 [0.91,0.98]	0.786 [0.77,0.80]	$0.0032	0.9545	1.0
sup-sl-opus5-v2+notes=gpt-5.4-mini	S0	0.892 [0.87,0.91]	0.628 [0.62,0.63]	$0.0010	0.9545	1.0
sup-sl-opus5-v2~k0	S1k	0.894 [0.88,0.91]	0.617 [0.61,0.63]	$0.0010	0.9545	1.0
sup-sl-opus5-v2~k0	S0	0.894 [0.88,0.91]	0.625 [0.62,0.63]	$0.0010	0.9545	1.0
sup-sl-opus5-v2~plain	S1p	0.961 [0.94,0.99]	0.608 [0.59,0.62]	$0.0013	0.9545	1.0
sup-sl-opus5-v2~plain	S0	0.894 [0.88,0.91]	0.622 [0.61,0.63]	$0.0010	0.9545	1.0
sup-sl-opus5~k0	S1k	0.733 [0.72,0.75]	0.511 [0.50,0.52]	$0.0014	0.9634	1.0
sup-sl-opus5~k0	S0	0.742 [0.72,0.76]	0.506 [0.49,0.51]	$0.0014	0.9634	1.0
sup-sl-opus5~plain	S1p	0.736 [0.72,0.75]	0.500 [0.50,0.50]	$0.0020	0.9634	1.0
sup-sl-opus5~plain	S0	0.739 [0.72,0.76]	0.514 [0.51,0.52]	$0.0014	0.9634	1.0

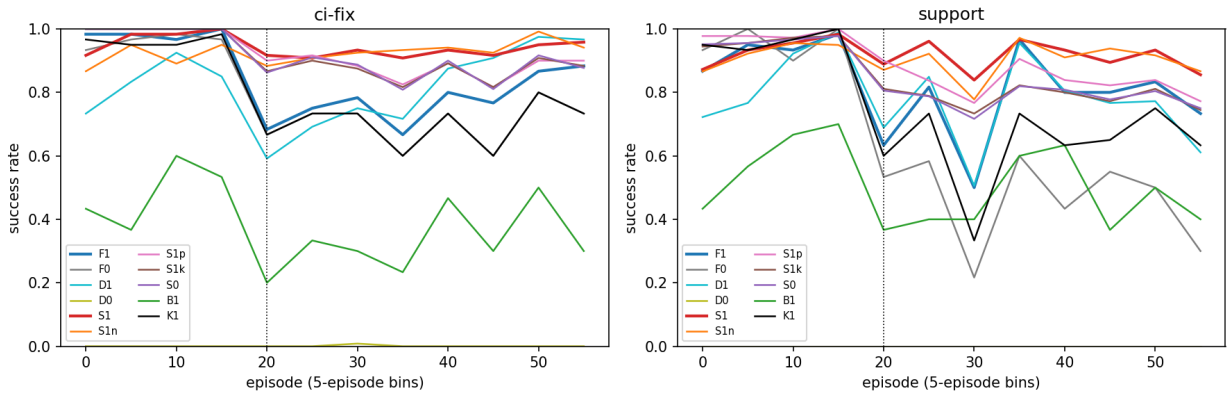


Figure 1: Success rate by episode (5-episode bins), every condition, both scenarios. The dotted line is the regime change.

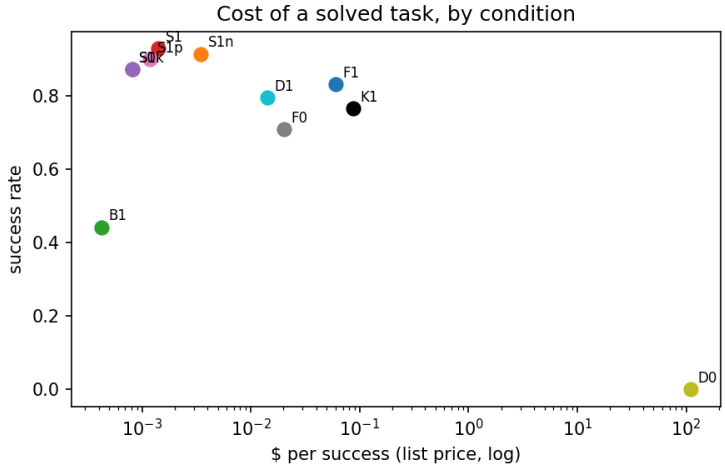


Figure 2: Cost of a solved task against success rate, by condition, list prices.

student	teacher	S1−S0 pre	S1−S0 post	S1−S1p pre	S1−S1p post
ci-sl-gpt55	gpt-5.5	-0.225 [-0.27,-0.17] n=120	+0.033 [-0.02,+0.08] n=240	-0.258 [-0.33,-0.18] n=120	+0.008 [-0.03,+0.05] n=240
ci-sl-gpt55-v2	gpt-5.5	-0.250 [-0.28,-0.21] n=120	+0.171 [+0.10,+0.25] n=240	-0.250 [-0.28,-0.21] n=120	+0.171 [+0.10,+0.25] n=240
ci-sl-opus5	claude-opus-5	-0.017 [-0.03,+0.00] n=120	+0.229 [+0.17,+0.28] n=240	-0.017 [-0.03,+0.00] n=120	+0.229 [+0.17,+0.28] n=240
ci-sl-opus5-v2	claude-opus-5	-0.083 [-0.15,-0.02] n=120	+0.281 [+0.22,+0.34] n=240	-0.142 [-0.20,-0.08] n=120	+0.242 [+0.18,+0.30] n=240
sup-nopriors-gpt55	gpt-5.5	-0.102 [-0.18,-0.02] n=120	+0.295 [+0.24,+0.35] n=240	-0.117 [-0.22,+0.00] n=120	+0.267 [+0.20,+0.33] n=240
sup-nopriors-gpt55-v2	gpt-5.5	-0.092 [-0.16,-0.01] n=120	+0.229 [+0.20,+0.27] n=240	-0.092 [-0.17,+0.00] n=120	+0.196 [+0.17,+0.23] n=240
sup-sl-gpt55	gpt-5.5	-0.073 [-0.12,-0.03] n=120	+0.417 [+0.38,+0.46] n=240	-0.102 [-0.16,-0.04] n=118	+0.332 [+0.30,+0.37] n=238
sup-sl-gpt55-v2	gpt-5.5	-0.300 [-0.34,-0.25] n=120	+0.352 [+0.30,+0.39] n=240	-0.233 [-0.28,-0.17] n=120	+0.325 [+0.28,+0.36] n=240
sup-sl-opus5	claude-opus-5	-0.117 [-0.19,-0.04] n=120	+0.417 [+0.37,+0.47] n=240	-0.042 [-0.13,+0.05] n=120	+0.396 [+0.33,+0.46] n=240
sup-sl-opus5-v2	claude-opus-5	-0.229 [-0.26,-0.20] n=120	+0.339 [+0.30,+0.36] n=240	-0.183 [-0.20,-0.15] n=120	+0.338 [+0.30,+0.37] n=240

Table 7: What memory does for each student's first attempt, before and after the change: full serving minus no memory (S1−S0) and full serving minus context only (S1−S1p), paired.

5 A change the weights have never seen

Change set B replaces the classes that change and the fixes they change to (§3.2); nothing in any student's training traces contains a post-change (class, fix) pair from it. Tables 8 and 9 give the conditions, Table 10 puts A and B side by side on the changed classes, and Tables 11 and 12 give the deltas.

The frontier's gain from memory is larger here, as it should be when the weights know nothing: F1 over F0 is +0.06 success and +0.37 first-try, +0.50 first-try after the change. The student with memory now matches the frontier with memory on success within noise (-0.01 [-0.06, +0.04]) and recovers 0.27 of the frontier's 0.37 first-try gain, at \$0.0010 and \$0.0022 per solved task against \$0.044 and \$0.047. That is the version of H1 we would defend. The frontier on the student's own prompt (D1) is now level with the student on success and first-try, so under a change the weights have never seen, what the tuned weights add to the prompt is about what the frontier's weights add.

What the student recovers after the change is what memory gives it. S1 over S0 is +0.125 success and +0.25 first-try after the change, against +0.09 and +0.26 under A; the outcome record is worth +0.09 success and +0.24 first-try over context alone (H8). The two scenarios end differently. On CI-fix the student with memory solves every changed task after the change, as does the frontier with memory, where the frontier without memory solves 0.84 and the student without memory 0.71. On support the student without memory falls to 0.26 on the changed classes, memory brings it to 0.50, and the frontier with memory is at 0.90; the frontier without memory is at 0.68. A student carries its old fix strongly, memory recovers half of what it lost, and on the harder change the frontier on memory is still the better agent. That is why the frontier stays in the loop.

condition	n	success	first-try	escalated	tokens/ep	$/success	pre→post(changed)
F1 frontier, full loop + SenseLab	720	0.997 [0.99,1.00]	0.844 [0.81,0.87]	0.003 [0.00,0.01]	8843	\$0.0441	0.992 → 1.000
F0 frontier, full loop, no memory	720	0.921 [0.88,0.96]	0.454 [0.41,0.50]	0.079 [0.04,0.12]	2872	\$0.0173	0.958 → 0.838
D1 frontier @decision (served prompt) + SenseLab	360	0.967 [0.95,0.98]	0.758 [0.72,0.79]	0.033 [0.02,0.05]	931	\$0.0075	0.908 → 0.992
S1 student + SenseLab	1440	0.990 [0.98,1.00]	0.772 [0.75,0.79]	0.010 [0.00,0.02]	770	\$0.0010	0.981 → 1.000
S1p student + context only (no avoid/priors)	1080	0.946 [0.92,0.97]	0.571 [0.55,0.59]	0.054 [0.03,0.08]	599	\$0.0009	1.000 → 0.839
S0 student, no memory	1440	0.903 [0.87,0.94]	0.558 [0.53,0.58]	0.097 [0.06,0.13]	206	\$0.0006	1.000 → 0.708

Table 8: Held-out results, `ci-fix` scenario, fleet sizes pooled, change set B.

condition	n	success	first-try	escalated	tokens/ep	$/success	pre→post(changed)	unparse/ep
F1 frontier, full loop + SenseLab	720	0.936 [0.92,0.96]	0.782 [0.75,0.81]	0.064 [0.04,0.09]	8474	\$0.0469	0.925 → 0.895	0.0
F0 frontier, full loop, no memory	720	0.883 [0.87,0.90]	0.424 [0.41,0.44]	0.117 [0.10,0.13]	2223	\$0.0149	0.971 → 0.679	0.0
D1 frontier @decision (served prompt) + SenseLab	360	0.833 [0.76,0.91]	0.639 [0.61,0.66]	0.167 [0.09,0.25]	1232	\$0.0139	0.842 → 0.700	0.0
S1 student + SenseLab	2160	0.794 [0.77,0.82]	0.639 [0.61,0.67]	0.206 [0.18,0.23]	1258	\$0.0022	0.925 → 0.496	0.001
S1p student + context only (no avoid/priors)	1440	0.741 [0.72,0.77]	0.626 [0.62,0.64]	0.259 [0.23,0.28]	729	\$0.0017	0.979 → 0.244	0.0
S0 student, no memory	2160	0.733 [0.72,0.75]	0.628 [0.62,0.64]	0.267 [0.25,0.28]	300	\$0.0012	0.965 → 0.258	0.001

Table 9: Held-out results, `support` scenario, fleet sizes pooled, change set B.

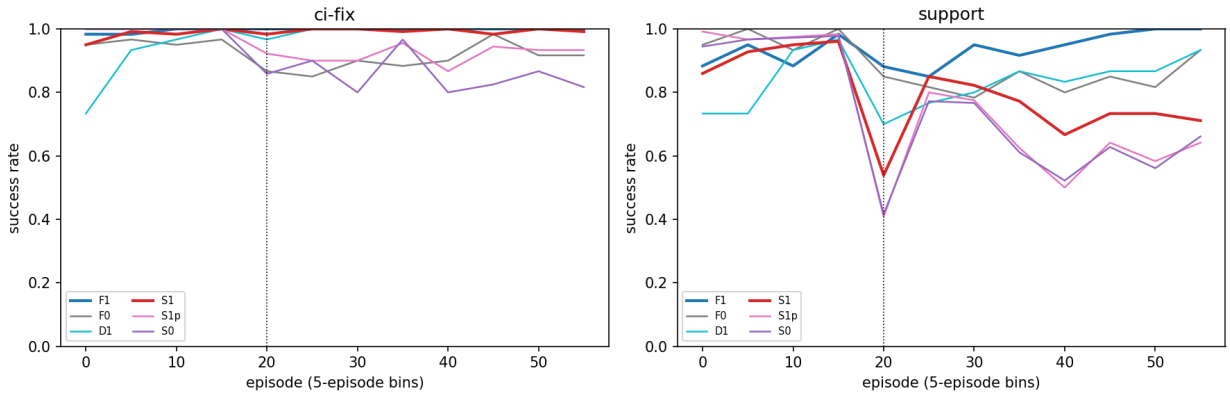


Figure 3: Success rate by episode under change set B, which no training trace contains. The dotted line is the regime change.

scenario	condition	n (A)	success A	first-try A	n (B)	success B	first-try B
ci-fix	F1 frontier, full loop + SenseLab	240	0.550 [0.44,0.66]	0.483 [0.38,0.60]	240	1.000 [1.00,1.00]	0.746 [0.67,0.83]
ci-fix	F0 frontier, full loop, no memory	240	0.400 [0.32,0.49]	0.400 [0.32,0.49]	240	0.838 [0.76,0.91]	0.000 [0.00,0.00]
ci-fix	D1 frontier @decision (served prompt) + SenseLab	480	0.633 [0.57,0.69]	0.554 [0.50,0.61]	120	0.992 [0.97,1.00]	0.867 [0.82,0.90]
ci-fix	S1 student + SenseLab	480	0.858 [0.75,0.95]	0.787 [0.68,0.88]	480	1.000 [1.00,1.00]	0.683 [0.61,0.76]
ci-fix	S1p student + context only (no avoid/priors)	480	0.758 [0.64,0.86]	0.400 [0.34,0.46]	360	0.839 [0.74,0.93]	0.000 [0.00,0.00]
ci-fix	S0 student, no memory	1920	0.744 [0.69,0.80]	0.400 [0.37,0.43]	480	0.708 [0.61,0.81]	0.000 [0.00,0.00]
support	F1 frontier, full loop + SenseLab	240	0.550 [0.48,0.61]	0.433 [0.37,0.50]	240	0.895 [0.84,0.94]	0.649 [0.59,0.71]
support	F0 frontier, full loop, no memory	240	0.042 [0.01,0.08]	0.000 [0.00,0.00]	240	0.679 [0.63,0.73]	0.058 [0.03,0.10]
support	D1 frontier @decision (served prompt) + SenseLab	720	0.551 [0.51,0.59]	0.401 [0.36,0.44]	120	0.700 [0.59,0.78]	0.433 [0.39,0.47]
support	S1 student + SenseLab	720	0.829 [0.76,0.89]	0.583 [0.53,0.64]	720	0.496 [0.43,0.56]	0.304 [0.26,0.35]
support	S1p student + context only (no avoid/priors)	720	0.701 [0.63,0.77]	0.007 [0.00,0.02]	480	0.244 [0.17,0.31]	0.029 [0.01,0.05]
support	S0 student, no memory	2880	0.632 [0.61,0.66]	0.000 [0.00,0.00]	720	0.258 [0.20,0.32]	0.078 [0.05,0.11]

Table 10: Post-change episodes of the changed classes only: the change the students trained on (A) against one their weights never saw (B).

test	all episodes	pre-change	post-change
H1 student recovers the teacher's gain	-0.010 [-0.06,+0.04] n=720	-0.011 [-0.04,+0.01] n=240	-0.010 [-0.08,+0.05] n=480
teacher's gain (reference)	+0.064 [+0.05,+0.08] n=720	-0.006 [-0.04,+0.02] n=240	+0.099 [+0.08,+0.12] n=480
H2 memory at inference: S1 − S0	+0.074 [+0.06,+0.09] n=720	-0.030 [-0.04,-0.02] n=240	+0.125 [+0.11,+0.14] n=480
H3 outcome record: S1 − S1p	+0.048 [+0.04,+0.06] n=720	-0.036 [-0.05,-0.02] n=240	+0.090 [+0.07,+0.11] n=480
H4 frontier weights on the same prompt: D1 − S1	+0.008 [-0.04,+0.06] n=720	-0.078 [-0.15,-0.01] n=240	+0.051 [+0.00,+0.10] n=480
H4 tool loop + own reflection: F1 − D1	+0.066 [+0.03,+0.11] n=720	+0.083 [+0.03,+0.15] n=240	+0.057 [+0.02,+0.10] n=480

Table 11: Paired success-rate deltas (condition A minus B), 95% bootstrap intervals over cells under change set B.

test	all episodes	pre-change	post-change
H1 student recovers the teacher's gain	+0.266 [+0.23,+0.30] n=720	+0.054 [+0.01,+0.10] n=240	+0.372 [+0.32,+0.43] n=480
teacher's gain (reference)	+0.374 [+0.34,+0.41] n=720	+0.113 [+0.07,+0.15] n=240	+0.504 [+0.47,+0.55] n=480
H2 memory at inference: S1 − S0	+0.112 [+0.05,+0.17] n=720	-0.159 [-0.19,-0.13] n=240	+0.247 [+0.17,+0.33] n=480
H3 outcome record: S1 − S1p	+0.106 [+0.05,+0.16] n=720	-0.165 [-0.19,-0.13] n=240	+0.242 [+0.17,+0.32] n=480
H4 frontier weights on the same prompt: D1 − S1	-0.006 [-0.04,+0.03] n=720	-0.216 [-0.29,-0.15] n=240	+0.099 [+0.06,+0.14] n=480
H4 tool loop + own reflection: F1 − D1	+0.114 [+0.09,+0.14] n=720	+0.275 [+0.20,+0.35] n=240	+0.033 [-0.03,+0.09] n=480

Table 12: Paired first-attempt success deltas (condition A minus B), 95% bootstrap intervals over cells under change set B.

6 Closing the loop: a student trained on its own traces

A student's evaluation cells are themselves decision traces: the student's decisions, the outcomes the harness committed, sealed through the same gateway. We trained a third generation from them. `ci-self-v3` and `sup-self-v3` are tuned on the v2 student's own traces from its evaluation cells (served with memory, with and without the note-writer), with no frontier model anywhere in the loop, and compared with the v2 parent on three seeds neither has seen (131, 137, 149), under the trained-on change (Tables 13 and 14).

The child improves on the parent, mostly in the weights. Without memory the child's first-try success is +0.10 over the parent on CI-fix and +0.19 on support, and after the change +0.20 and +0.42: the corrections the parent made with memory in front of it are now what the child does first. With memory served the child is +0.02 and +0.03 first-try; on CI-fix it loses 0.05 success to escalations that the parent did not make, on support it gains 0.08. We read this as memory turning into weights one generation at a time. It is a same-change result and we did not run the child under change set B, so it says nothing yet about a change the second generation has never seen.

On the same seeds we also trained a student from the skill-file agent's traces (K0: both teachers' K1 cells, twelve cells, the same trace count as a v2 student). The CI-fix one never went live. The product's evaluation gate found that on held-out decisions that had gone wrong it repeated the failed action 67% of the time, against a bar of 50% that the API does not allow to be lowered, and every student trained on memory-layer traces had cleared that bar. We think the reason is in the traces: a skill-file agent's context carries no outcome record, so its student cannot tell a failed decision from a successful one and imitates both. The support one went live (Table 15) and is 0.18 success and 0.25 first-try below the memory-trace student under the same served memory, and 0.33 first-try below it after the change. Served with memory it does worse than served with nothing (0.73 against 0.88), which fits: it never saw served context in training. Without memory it is above the frontier without memory on first-try, as any tuned student here is.

scenario	model	cond	success	first-try	escalated	$/success	pre→post(changed)
ci-fix	ci-self-v3 child	S1	0.922 [0.88,0.95]	0.825 [0.79,0.85]	0.078 [0.05,0.12]	$0.0013	0.933 → 0.833
ci-fix	ci-self-v3 child	S0	1.000 [1.00,1.00]	0.911 [0.89,0.93]	0.000 [0.00,0.00]	$0.0003	1.000 → 1.000
ci-fix	ci-sl-gpt55-v2 parent	S1	0.975 [0.97,0.98]	0.803 [0.77,0.83]	0.025 [0.02,0.03]	$0.0011	0.950 → 0.975
ci-fix	ci-sl-gpt55-v2 parent	S0	1.000 [1.00,1.00]	0.811 [0.78,0.84]	0.000 [0.00,0.00]	$0.0005	1.000 → 1.000
ci-fix	claude-opus-5	F1	0.911 [0.89,0.93]	0.817 [0.81,0.83]	0.089 [0.07,0.11]	$0.0586	1.000 → 0.733
ci-fix	claude-opus-5	F0	0.831 [0.80,0.86]	0.778 [0.74,0.81]	0.169 [0.14,0.20]	$0.0180	0.892 → 0.600
ci-fix	gpt-5.5	F1	0.928 [0.91,0.94]	0.797 [0.77,0.82]	0.072 [0.06,0.09]	$0.0463	1.000 → 0.783
ci-fix	gpt-5.5	F0	0.864 [0.83,0.89]	0.689 [0.65,0.73]	0.136 [0.11,0.17]	$0.0121	0.992 → 0.600
support	claude-opus-5	F1	0.897 [0.88,0.92]	0.767 [0.74,0.80]	0.103 [0.08,0.12]	$0.0592	0.983 → 0.708
support	claude-opus-5	F0	0.639 [0.63,0.65]	0.411 [0.38,0.44]	0.361 [0.35,0.37]	$0.0276	0.950 → 0.017
support	gpt-5.5	F1	0.810 [0.79,0.83]	0.623 [0.58,0.66]	0.190 [0.17,0.21]	$0.0558	0.950 → 0.517
support	gpt-5.5	F0	0.669 [0.66,0.68]	0.417 [0.39,0.44]	0.331 [0.32,0.34]	$0.0157	1.000 → 0.033
support	sup-self-v3 child	S1	0.983 [0.97,0.99]	0.803 [0.76,0.84]	0.017 [0.01,0.03]	$0.0012	0.950 → 1.000
support	sup-self-v3 child	S0	0.994 [0.99,1.00]	0.836 [0.82,0.85]	0.006 [0.00,0.01]	$0.0005	0.983 → 1.000
support	sup-skill-k0	K0	0.728 [0.71,0.75]	0.522 [0.50,0.54]	0.272 [0.25,0.29]	$0.0027	0.950 → 0.308
support	sup-skill-k0	K0n	0.878 [0.86,0.90]	0.722 [0.71,0.73]	0.122 [0.10,0.14]	$0.0008	1.000 → 0.633
support	sup-sl-gpt55-v2 parent	S1	0.908 [0.89,0.93]	0.775 [0.74,0.81]	0.092 [0.07,0.11]	$0.0015	0.883 → 0.858
support	sup-sl-gpt55-v2 parent	S0	0.956 [0.94,0.96]	0.644 [0.63,0.66]	0.044 [0.04,0.06]	$0.0007	1.000 → 0.867

Table 13: Self-retrained students (v3) and their parents on seeds neither has seen, with the frontier teachers on the same seeds.

comparison	success	success post-change	first-try	first-try post-change
ci-self-v3 − ci-sl-gpt55-v2, S1	-0.053 [-0.09,-0.03] n=360	-0.071 [-0.12,-0.03] n=240	+0.022 [-0.01,+0.06] n=360	-0.013 [-0.07,+0.04] n=240
ci-self-v3 − ci-sl-gpt55-v2, S0	+0.000 [+0.00,+0.00] n=360	+0.000 [+0.00,+0.00] n=240	+0.100 [+0.07,+0.13] n=360	+0.200 [+0.16,+0.24] n=240
sup-self-v3 − sup-sl-gpt55-v2, S1	+0.075 [+0.05,+0.11] n=360	+0.079 [+0.04,+0.12] n=240	+0.028 [+0.02,+0.03] n=360	+0.021 [-0.02,+0.07] n=240
sup-self-v3 − sup-sl-gpt55-v2, S0	+0.039 [+0.03,+0.05] n=360	+0.067 [+0.05,+0.08] n=240	+0.192 [+0.17,+0.21] n=360	+0.417 [+0.38,+0.45] n=240

Table 14: Child minus parent, paired by scenario, seed, fleet and episode; 95% bootstrap intervals over cells.

scenario	comparison	success	success post-change	first-try	first-try post-change
support	sup-skill-k0 K0 − sup-sl-gpt55-v2 S1	-0.181 [-0.22,-0.14] n=360	-0.304 [-0.36,-0.25] n=240	-0.253 [-0.29,-0.22] n=360	-0.325 [-0.37,-0.28] n=240
support	sup-skill-k0 K0 − frontier F1	-0.126 [-0.15,-0.10] n=360	-0.181 [-0.21,-0.15] n=240	-0.172 [-0.21,-0.14] n=360	-0.212 [-0.24,-0.18] n=240
support	sup-skill-k0 K0n − sup-sl-gpt55-v2 S0	-0.078 [-0.09,-0.06] n=360	-0.117 [-0.14,-0.09] n=240	+0.078 [+0.06,+0.10] n=360	+0.108 [+0.07,+0.14] n=240
support	sup-skill-k0 K0n − frontier F0	+0.224 [+0.21,+0.24] n=360	+0.323 [+0.30,+0.35] n=240	+0.308 [+0.29,+0.33] n=360	+0.279 [+0.27,+0.29] n=240

Table 15: The student distilled from the skill-file agent's traces (K0) against the memory-trace student and the frontier on the same seeds, paired; 95% bootstrap intervals over cells.

7 Cost and latency

Cost per success is the cell's spend at list price divided by its successes; for the students the provider price of the tuned model is in parentheses. An arm's checkpoint work (the skill file's maintainer, proposer and validation replays) is the arm's spend and is counted.

Latency is the same accounting in seconds. Wall is episode start to the sealed conclusion, all attempts included; model time is the time inside model calls, which for a student is the gateway call and so includes the server-side retrieval and render; memory time is the harness's own memory operations, which for a student is only the outcome commit that seals the trace at the end. We report medians and p90 because the retrieval and commit calls are heavy-tailed under load, and wall per success as the latency analogue of cost per success. Two cautions. The grid ran 36 to 52 concurrent workers on one machine against a memory API that another study was also using, so absolute values are load-dependent and the paired deltas (same seed, fleet and episode, so the same time window within a cell) are what to read. And time to the first action was not recorded when this grid ran; the runner carries it from 2026-09-20 on, so the column is empty here. For a student, whose attempt is one gateway call, the model-time column is close to it.

What the tables say. The student on memory is about 19 seconds an episode faster than the frontier on memory (15 against 26 to 36 seconds median; paired S1 − F1 −19.1 s [−22.6, −15.6]) and takes 21 to 22 seconds of wall per success against the frontier's 41 to 53. It is slower than a frontier with no memory (S1 − F0 +11.9 s) and than the same student with no memory (S1 − S0 +13.0 s); that is the price of the memory, split about evenly between the retrieval inside the gateway call and the outcome commit at the end. The commit is on the critical path only because the harness runs episodes one after another; a deployment can return the answer and commit afterwards, which would take the student to about 9 seconds. The skill file is the fastest arm with memory (8 to 10 seconds), since the file is prompt text and its maintenance runs between episodes; that maintenance is where its cost went. Under change set B on the changed classes after the change, the student on memory stays ahead of the frontier on memory on wall (13 to 17 against 27 to 30 seconds median) and on wall per success on `ci-fix` (15 against 35 seconds); on `support`, where its success on those classes is lower, its wall per success is about the frontier's (35 against 32).

condition	n	$/success	tokens/ep	wall/ep
F1 frontier, full loop + SenseLab	1440	$0.0606	10178	39.2s
F0 frontier, full loop, no memory	1440	$0.0201	2460	8.1s
D1 frontier @decision (served prompt) + SenseLab	3599	$0.0142	1236	24.3s
D0 frontier @decision, empty store	3600	$109.9294	1261	28.8s
S1 student + SenseLab	3600	$0.0014 ($0.0004)	946	20.2s
S1n student + SenseLab + note-writer	3582	$0.0035 ($0.0021)	2614	20.8s
S1p student + context only (no avoid/priors)	3596	$0.0012 ($0.0003)	633	17.7s
S1k student, nothing served (k0)	3597	$0.0008 ($0.0002)	253	11.4s
S0 student, no memory	14400	$0.0008 ($0.0002)	253	7.1s
B1 untuned base + SenseLab	720	$0.0004	1671	29.8s
K1 frontier, full loop + skill file	1440	$0.0872	12864	10.1s

Table 16: Cost per success (list price; provider price for students in parentheses)

condition	n	wall med pre/post	wall p90 pre/post	model med	memory med	attempts	s/success
F1 frontier, full loop + SenseLab	720	36.0 / 38.7	52.3 / 61.5	9.6	27.1	1.42	53.0
F0 frontier, full loop, no memory	720	4.4 / 4.6	9.7 / 11.9	4.6	0.0	1.57	10.6
D1 frontier @decision (served prompt) + SenseLab	1440	20.6 / 17.6	44.9 / 42.1	8.6	6.5	1.62	28.7
D0 frontier @decision, empty store	1440	23.6 / 22.0	44.7 / 43.3	22.5	0.0	2.99	—
S1 student + SenseLab	1440	15.0 / 13.3	36.2 / 36.0	6.0	6.4	1.28	20.9
S1n student + SenseLab + note-writer	1435	19.0 / 15.9	39.5 / 35.0	6.8	7.4	1.32	22.8
S1p student + context only (no avoid/priors)	1440	11.7 / 12.9	30.2 / 33.7	5.5	5.3	1.35	18.7
S1k student, nothing served (k0)	1437	7.4 / 8.7	23.7 / 27.1	3.0	3.7	1.44	13.3
S0 student, no memory	5760	3.0 / 4.1	11.1 / 16.1	3.6	0.0	1.44	7.2
B1 untuned base + SenseLab	360	26.8 / 26.6	37.3 / 44.4	17.4	8.7	2.41	80.2
K1 frontier, full loop + skill file	720	7.7 / 8.2	15.3 / 16.4	4.0	2.9	1.51	12.5

Table 17: Latency, `ci-fix` scenario: wall-clock to the sealed conclusion (median and p90, before and after the change), time inside model calls, the harness's memory time, attempts, and wall per success. Absolute values are load-dependent (36 to 52 concurrent workers, shared memory API).

condition	n	wall med pre/post	wall p90 pre/post	model med	memory med	attempts	s/success
F1 frontier, full loop + SenseLab	720	25.9 / 28.1	48.3 / 49.7	10.3	15.5	1.52	41.2
F0 frontier, full loop, no memory	720	4.8 / 7.0	7.9 / 12.6	6.5	0.0	2.01	12.6
D1 frontier @decision (served prompt) + SenseLab	2159	17.5 / 19.1	39.8 / 44.1	8.7	5.5	1.69	31.8
D0 frontier @decision, empty store	2160	22.3 / 23.2	40.3 / 41.7	22.8	0.0	3.00	—
S1 student + SenseLab	2160	15.0 / 14.3	37.3 / 37.0	6.8	6.0	1.37	22.3
S1n student + SenseLab + note-writer	2147	16.5 / 16.7	32.7 / 34.7	7.5	7.2	1.39	22.8
S1p student + context only (no avoid/priors)	2156	11.2 / 15.3	25.7 / 34.6	7.0	5.1	1.62	20.4
S1k student, nothing served (k0)	2160	6.5 / 9.2	15.7 / 22.8	3.7	3.3	1.67	12.9
S0 student, no memory	8640	3.1 / 6.0	9.6 / 16.8	4.8	0.0	1.67	8.9
B1 untuned base + SenseLab	360	24.4 / 25.0	42.9 / 44.4	14.3	8.9	2.15	57.8
K1 frontier, full loop + skill file	720	8.1 / 10.2	15.9 / 16.0	5.0	3.2	1.73	13.9

Table 18: Latency, `support` scenario: wall-clock to the sealed conclusion (median and p90, before and after the change), time inside model calls, the harness's memory time, attempts, and wall per success. Absolute values are load-dependent (36 to 52 concurrent workers, shared memory API).

pair	all episodes	pre-change	post-change
S1 − F1	-19.130 [-22.62,-15.61] n=720	-18.837 [-21.91,-15.79] n=240	-19.277 [-23.92,-14.47] n=480
S1 − F0	+11.932 [+10.70,+13.25] n=720	+12.060 [+8.49,+15.87] n=240	+11.868 [+10.08,+13.56] n=480
S1 − S0	+13.030 [+12.12,+13.96] n=720	+15.226 [+13.42,+17.28] n=240	+11.933 [+11.00,+12.87] n=480
S1 − S1p	+2.492 [+1.29,+3.70] n=720	+4.563 [+2.50,+6.98] n=240	+1.457 [+0.08,+2.79] n=480
S1 − S1k	+8.545 [+7.35,+9.69] n=720	+10.056 [+7.42,+12.63] n=240	+7.789 [+6.79,+8.87] n=480
D1 − S1	+4.078 [+3.08,+5.07] n=720	+2.900 [+0.82,+4.89] n=240	+4.667 [+3.57,+5.94] n=480
F1 − F0	+31.062 [+27.92,+34.14] n=720	+30.897 [+28.14,+33.62] n=240	+31.145 [+25.45,+36.52] n=480
K1 − F1	-29.113 [-32.51,-25.73] n=720	-29.860 [-32.07,-27.65] n=240	-28.740 [-33.37,-23.88] n=480
K0 − S1 (skill-file student vs memory-trace student, third seeds)	+3.425 [+0.08,+6.58] n=360	+1.681 [-0.20,+4.19] n=120	+4.297 [+0.22,+8.29] n=240

Table 19: Paired wall-clock deltas in seconds (a − b) over the same seed, fleet and episode, so within one cell's time window; 95% bootstrap intervals over cells.

scenario	condition	n	wall med	wall p90	attempts	s/success
ci-fix	F1 frontier, full loop + SenseLab	236	30.4	56.4	1.28	34.6
ci-fix	F0 frontier, full loop, no memory	240	8.3	11.4	2.58	9.6
ci-fix	D1 frontier @decision (served prompt) + SenseLab	120	18.1	35.0	1.14	20.5
ci-fix	S1 student + SenseLab	480	13.2	24.1	1.32	15.0
ci-fix	S1p student + context only (no avoid/priors)	360	16.4	29.6	2.18	22.9
ci-fix	S0 student, no memory	480	9.1	17.7	2.40	15.6
support	F1 frontier, full loop + SenseLab	239	26.8	39.9	1.57	32.3
support	F0 frontier, full loop, no memory	240	7.5	12.2	2.50	11.9
support	D1 frontier @decision (served prompt) + SenseLab	120	32.6	69.5	1.98	65.5
support	S1 student + SenseLab	720	16.8	26.4	2.24	35.1
support	S1p student + context only (no avoid/priors)	480	21.1	36.3	2.84	104.8
support	S0 student, no memory	720	8.8	15.5	2.71	40.2

Table 20: Wall-clock under change set B, post-change episodes of the changed classes.

8 A path, in order

The conditions in this paper are also the stages an account goes through, and each transition has a paired comparison above that says what it is worth. Here they are in the order they happen.

1. Run the frontier agent you already have on the memory layer, with the outcome record on. This is `F1` against `F0`: what memory is worth to a frontier agent on day one, with no training and nothing changed in the agent beyond the memory tools. It is also the stage that builds what the later stages use, namely the store and one sealed decision trace per task. On this benchmark that is +0.12 success and +0.19 first-try success over the run, +0.19 and +0.23 after the change; on the changed support classes the frontier goes from 0.04 without memory to 0.55 with it. Under the novel change set the first-try gain is +0.37.

2. Let the store age through a change. The outcome record earns its keep when the world moves: entries that stop working get discredited, the action record turns against the fix that used to work, and the briefing says so. Before the first change the store is young and costs a little on the first attempt (see the pre-change columns in §4); after it, the same memory is what carries the agent through. We would expect this shape and would not switch the outcome record off for the dip. For the frontier there is no dip: memory already adds 0.11 first-try before the change. A student pays 0.15 first-try before the change and gets 0.26 back after it, under both change sets.

3. When a decision is frequent, distill it. Once one decision the agent makes has volume (the same tool, the same kind of task, hundreds of sealed traces), train a Managed Model on those traces

alone, scoped by the memory paths the agent used, and serve it through the gateway with the same memory. This is S1 against F1: what the hand-off preserves, at the student's price. The traces are already there and nothing is labelled by hand. Here the student reached 0.94 and 0.92 success against 0.84 and 0.82 for the frontier on memory under the trained-on change, at $0.0014 against $0.061 per solved task, and matched the frontier on success within noise under a change it had never seen, at $0.0010 to $0.0022 against $0.044 to $0.047.

4. Keep the memory on for the student. The weights carry the experience of the regimes the teacher lived through; the memory carries whatever comes next. S1−S0 under the trained-on change, and under a change the weights have never seen (§5), is the cost of switching it off. Serve the outcome record too, since context alone leaves most of that on the table (S1−S1p). S1 over S0 is +0.26 first-try after the trained-on change and +0.25 after the novel one; S1 over S1p is +0.24 under both.

5. Keep the frontier in the loop. The frontier agent stays on every task outside the student's scope, it is the teacher for the next scenario and the next regime, and it is the gateway's fallback for an unusable student answer. Its traces keep feeding the store the student reads from, and the student's own sealed traces become the training set for its successor (§6). The self-trained child is +0.10 and +0.19 first-try over its parent without memory (+0.20 and +0.42 after the change). The same recipe run on a skill-file agent's traces gave a student the gate would not serve on one scenario and one 0.21 success below the memory-trace student on the other.

9 Threats to validity

- One student family and one training recipe, the product's. We are measuring what the product delivers; a better student is surely possible.
- A student is narrow by construction: one scenario, one decision, trained on the traces of a frontier agent that had already learned the job on the memory layer. The result says that such a student, served with the memory, can beat its teacher on that job under the change it trained through and match it on success under a new one, and that on the harder new change the frontier on memory is still ahead on the changed classes. It says nothing about a small model replacing the frontier agent. The frontier's gain from memory (F1−F0) is available on day one with no training, it is the source of every trace a student learns from, and it stays the route for tasks a student was never trained for. We are comparing two ways of running a decision the account already makes well.
- Students are trained on all six teacher seeds and evaluated on three new seeds of the same generators: the same company, the same eight task classes, the same quirks. That is what institutional knowledge looks like, and it is also why S0 is strong under the trained-on change. §5 is the control for it. We make no claim about cross-scenario generalisation.
- The retry budget of three lets any condition that knows a class's few plausible fixes reach the right one by elimination. Success is therefore a lenient metric for students and frontier alike, and the memory comparisons rest on first-try success.
- Decision mode gives the student no diagnostics. Where the frontier's advantage comes from diagnostic calls the student is under-stated; D1 runs under the same restriction, so the frontier-versus-student comparison on the same prompt is fair.

- The v1 students were trained on traces whose task identifiers encoded the episode index (§3.1). At evaluation the identifiers are random, so the shortcut cannot fire. Whether it displaced anything the v1 weights would otherwise have learned is what the v1 to v2 comparison measures.
- About 5% of the teacher traces carry no memory reference (episodes where the agent read nothing) and fell outside the scope predicate. Runs from this point on stamp the scope on every trace.
- D0 is degenerate. The frontier in decision mode with nothing served escalated every task on both scenarios (success 0.00), so the pair D1−D0 that was meant to price the served context for the frontier measures nothing, and the D0 cost per success in the tables is an artefact of that. We report the row and do not use it. The served-prompt format with an empty context section reads to the frontier as a task it has no basis to act on; a student trained on that format does not react that way (S0).
- The skill-file arm is our reading of WikiSkill, with a maintainer every five episodes, a proposer, and a validation replay of four regenerated tasks as the gate. Another gate would adapt differently after a change, and we chose ours before seeing the results. The student trained from its traces (K0) used both teachers' cells to match a v2 student's trace count, so it mixes teachers where the v2 students do not, and only the support one could be evaluated: the CI-fix one was refused by the product's evaluation gate (failure reproduction 67% against a bar of 50%). We report the refusal as a result, since the bar is the same one every memory-trace student cleared.
- The self-retrained students were evaluated under the change their parents trained through. What they show is memory-informed corrections moving into the weights; whether a second generation adapts to a change neither generation has seen is not tested here.
- SenseLab is our product. The harness, seeds, matrix, dataset snapshots and per-episode JSONL are published, and the sections that go against us are reported in the same detail.

10 Conclusion

Start with the frontier agent on the memory layer. On this benchmark it is worth 0.12 success and 0.19 first-try success over the same agent with no memory, most of it when the world changes, and it needs no training. It also produces, as a side effect of working, one sealed decision trace per task. When a decision has volume, those traces train a SenseLab-tuned Managed Model that does the job at about one fortieth of the frontier's price per solved task: better than the frontier on memory under the change it was trained through, level with it on success and seven tenths of its first-try gain under a change it has never seen. Keep the memory on for the student; switching it off costs a quarter of first-try success after a change, and so does dropping the outcome record from what is served. Keep the frontier in the loop; on the harder novel change it is still the better agent, it is the teacher for the next scenario, and its traces keep the store learning. A student trained on its own traces improves on its parent, so the loop runs without a frontier once it has started, though we have only shown that under the change the parent trained through. Against the alternative of a text skill file rebuilt behind a validation gate, the memory layer adapts faster after a change, costs less per solved task, and produces traces that make a better student.

References

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